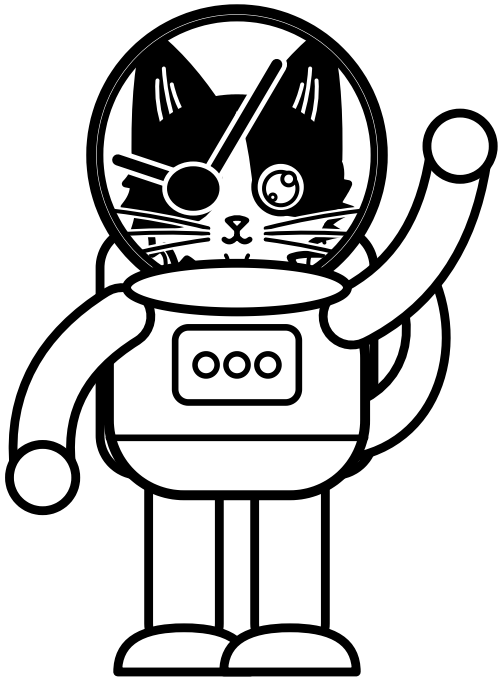
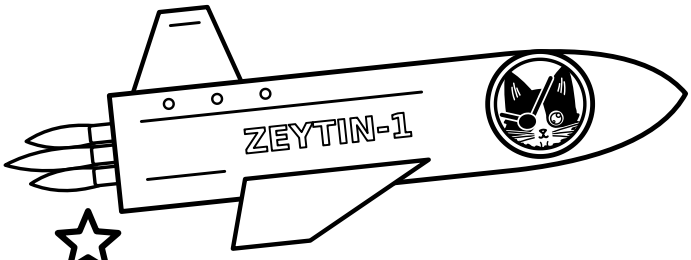
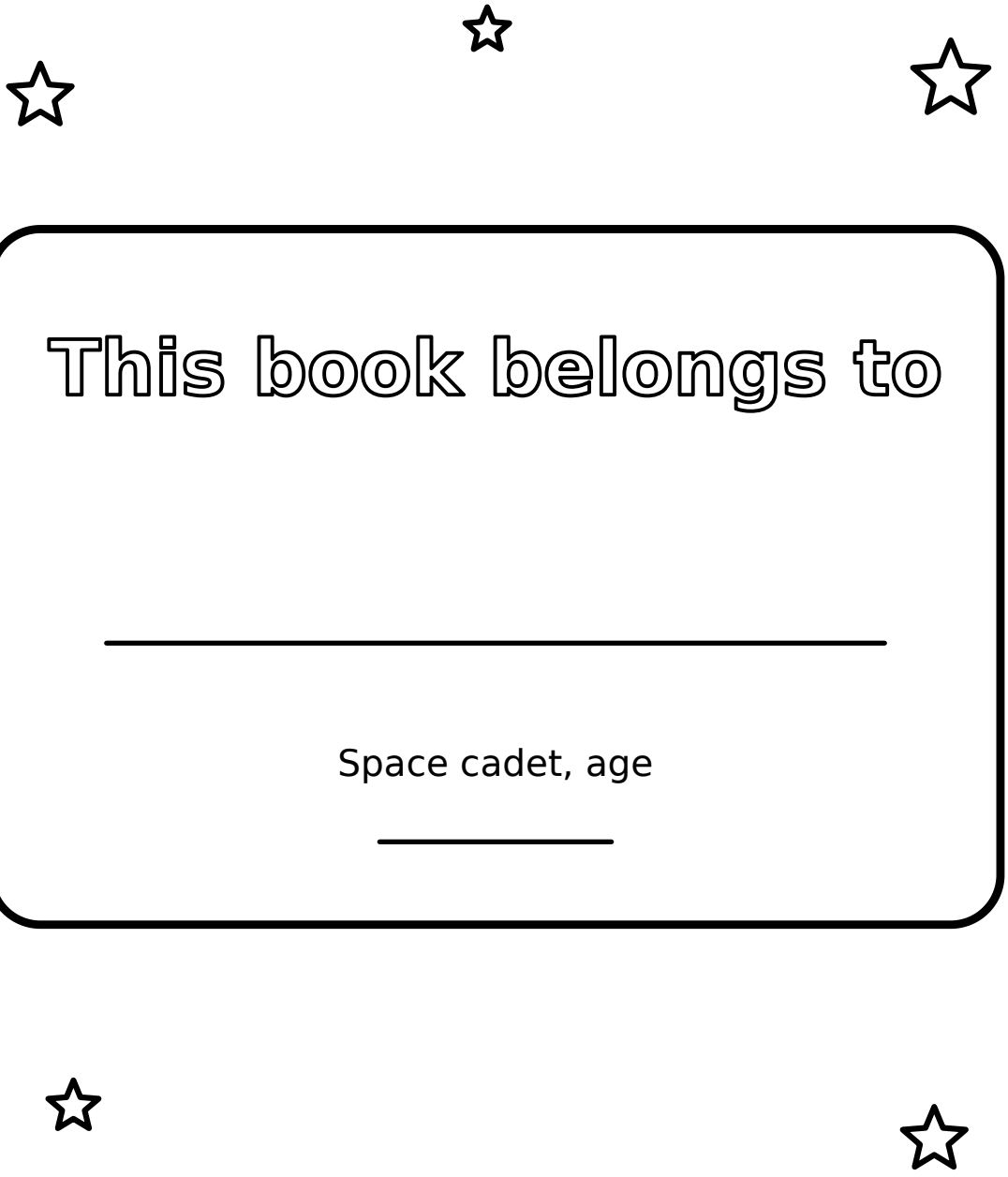
ZEYTIN

IN SPACE

A Coloring Journey Through the Solar System



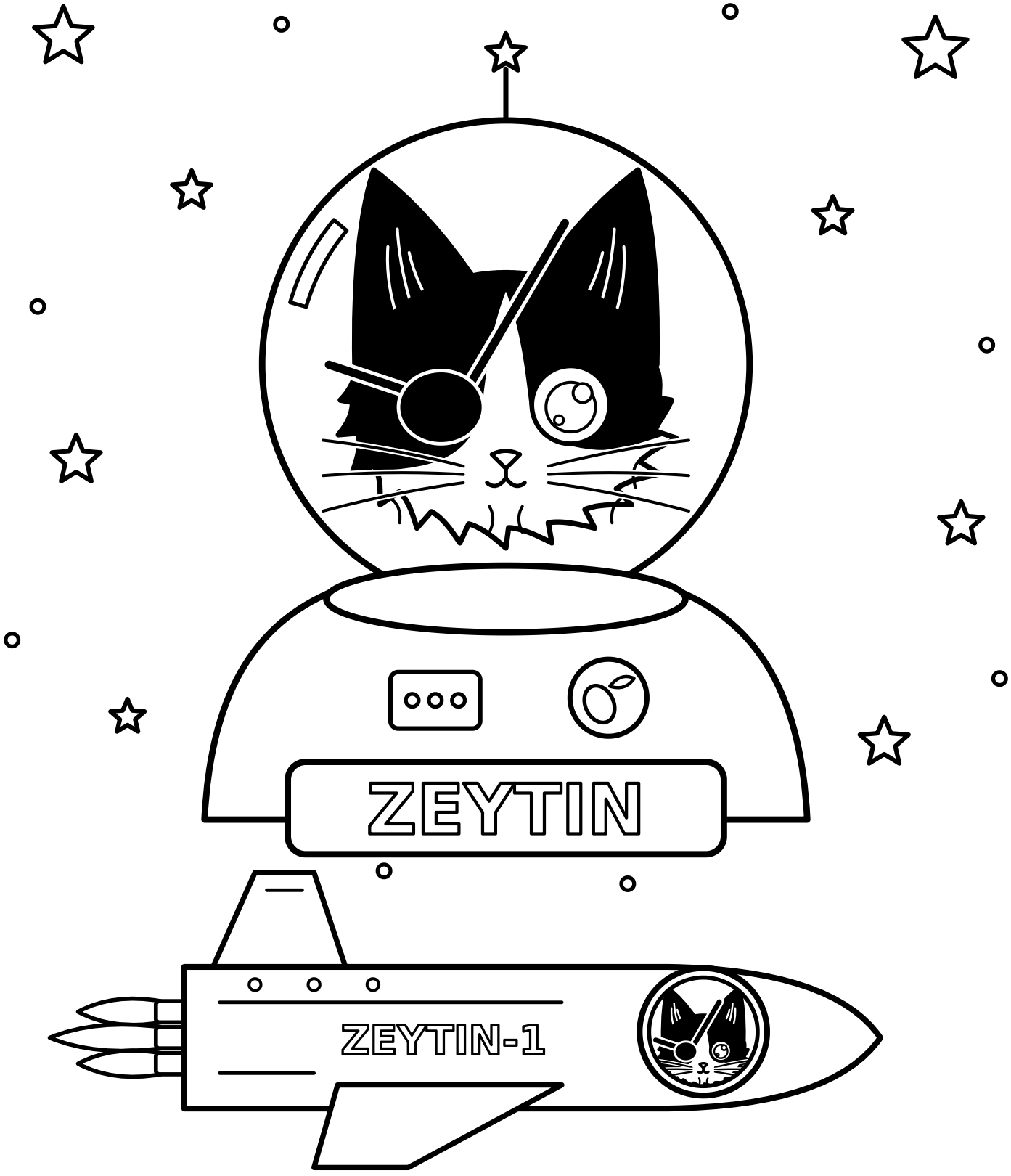




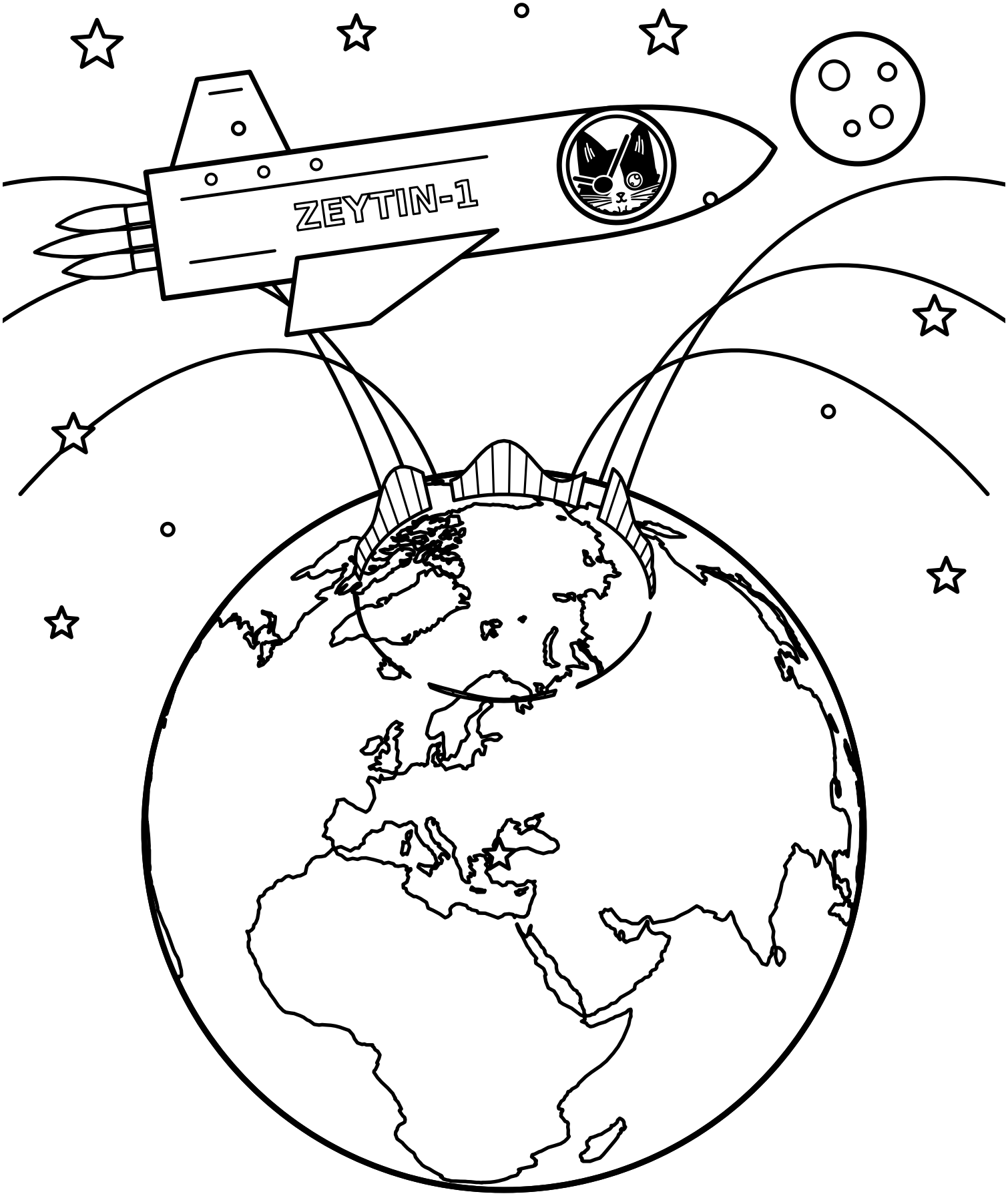
This book belongs to

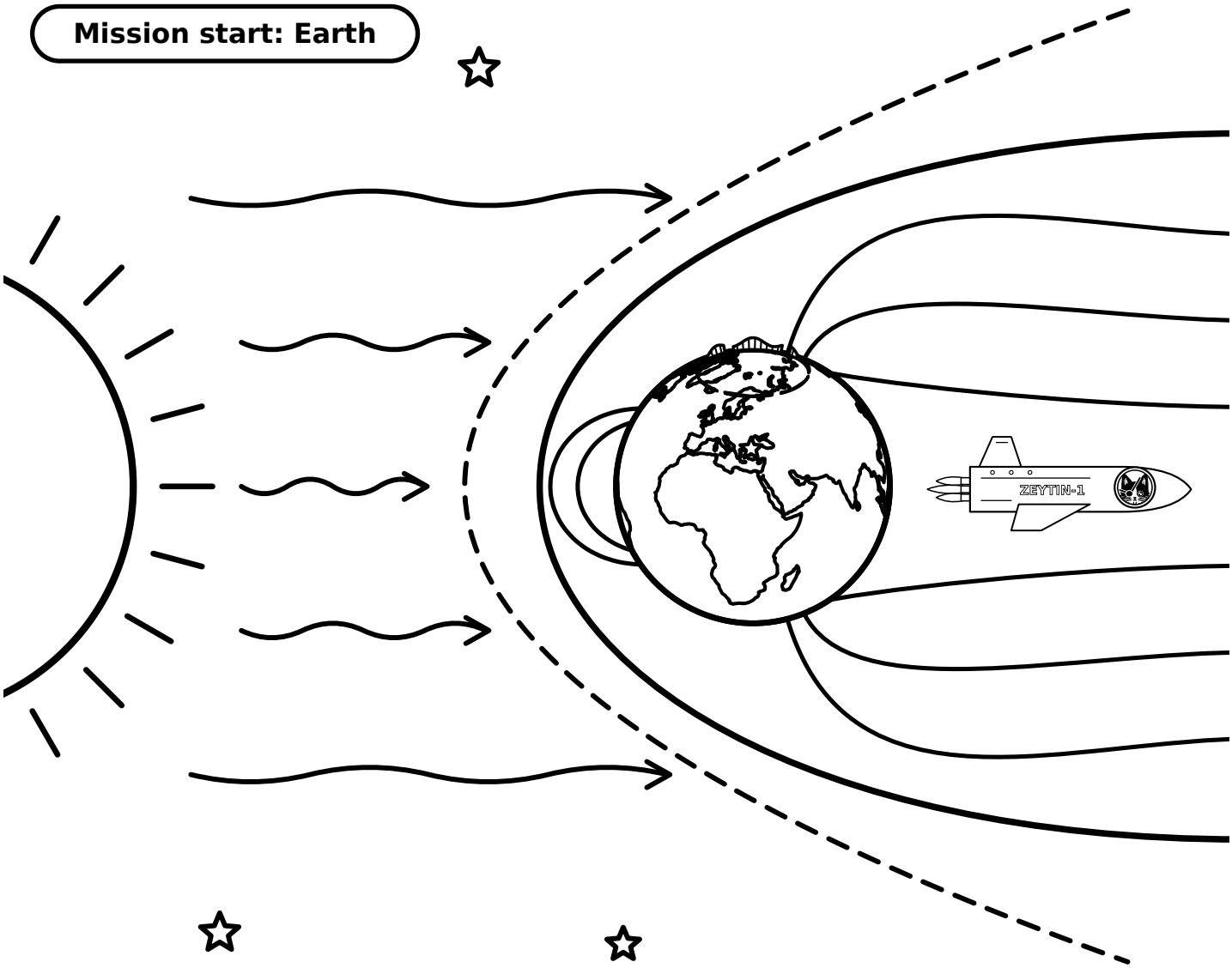
Space cadet, age

Tip: colored pencils and crayons work best;
markers may show through the paper.



Hi! I'm Zeytin, a fluffy space cat.
My name means “olive” in Turkish. Hop aboard my shuttle,
and let's explore the Solar System together!


ZEYTIN-1

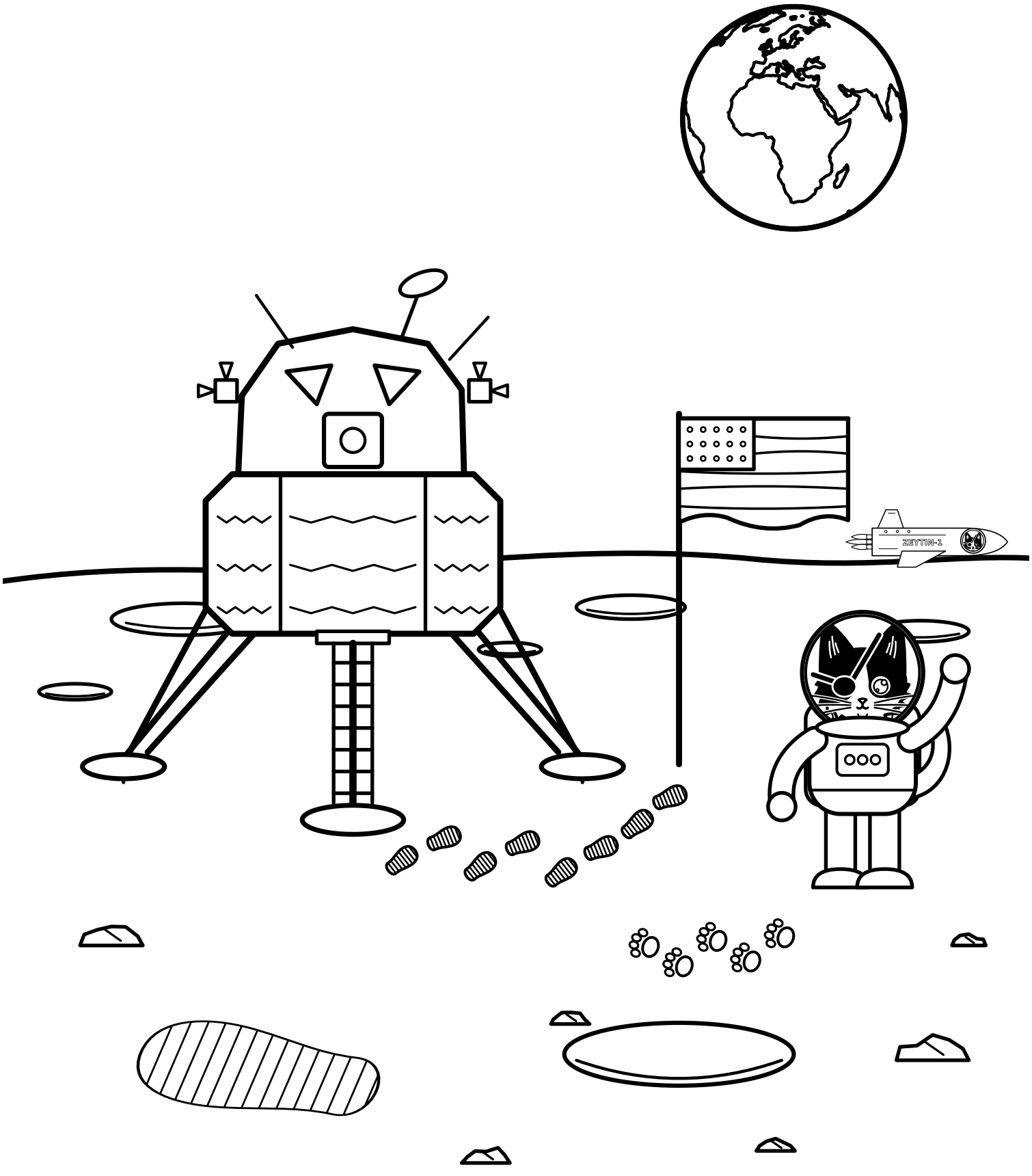


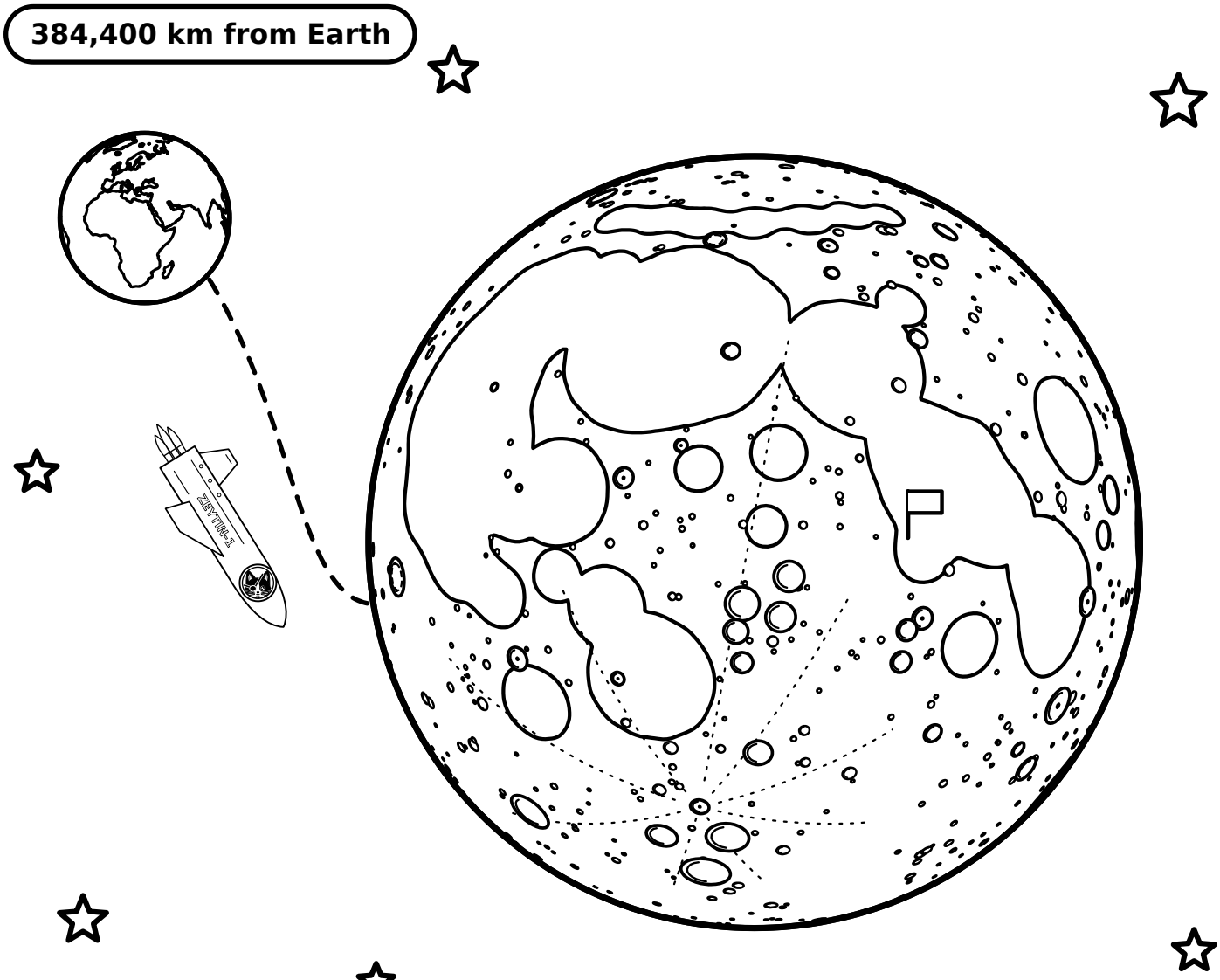
Earth’s Invisible Shield

Zeytin’s journey begins at home, on Earth, where a giant invisible magnetic shield surrounds our planet.

The shield pushes away the solar wind — a stream of tiny charged particles that rushes from the Sun at about 400 kilometers every second.

Near the North and South Poles, some particles slip in and make the air glow green and pink: the beautiful auroras!


ZEYTIN-1

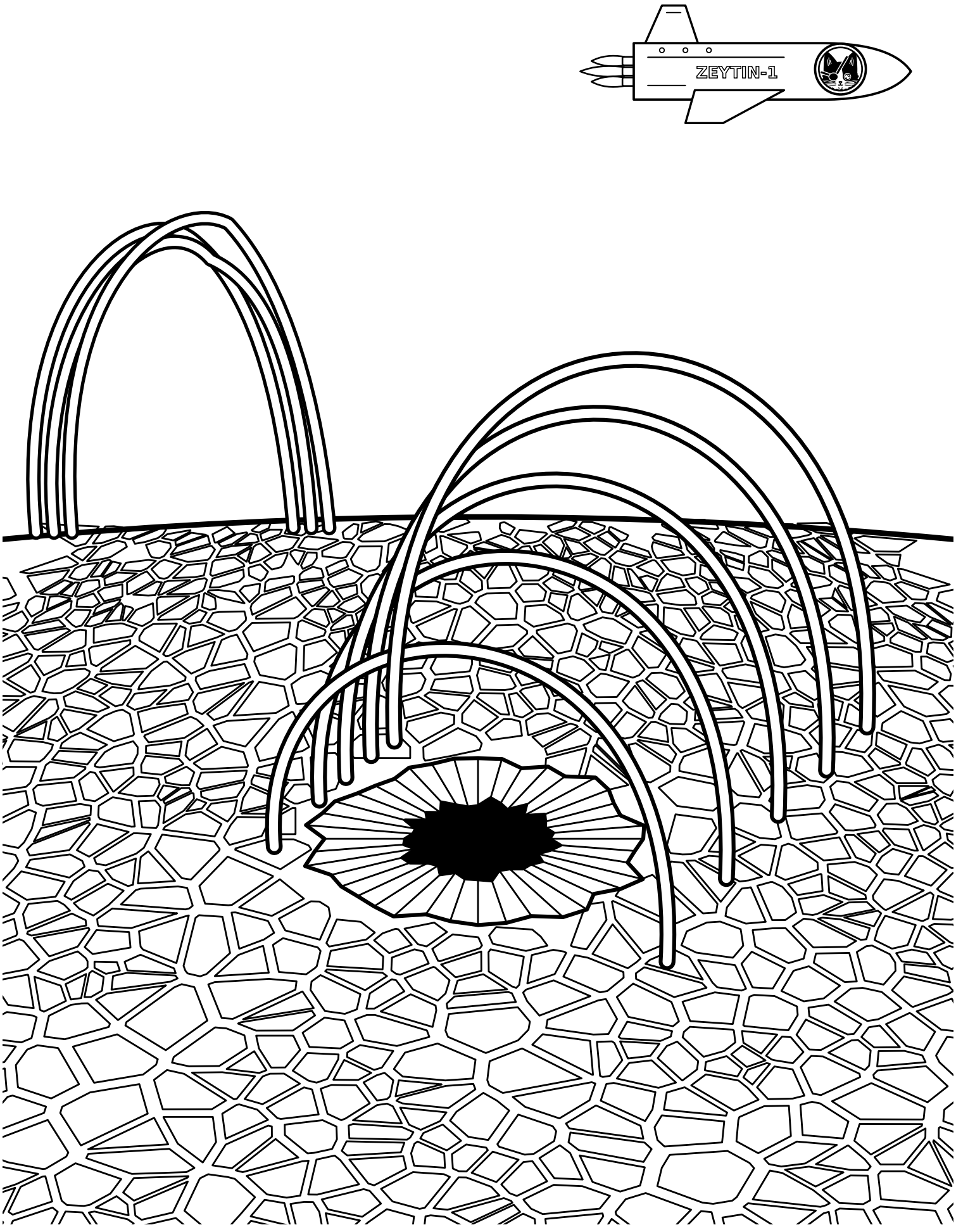


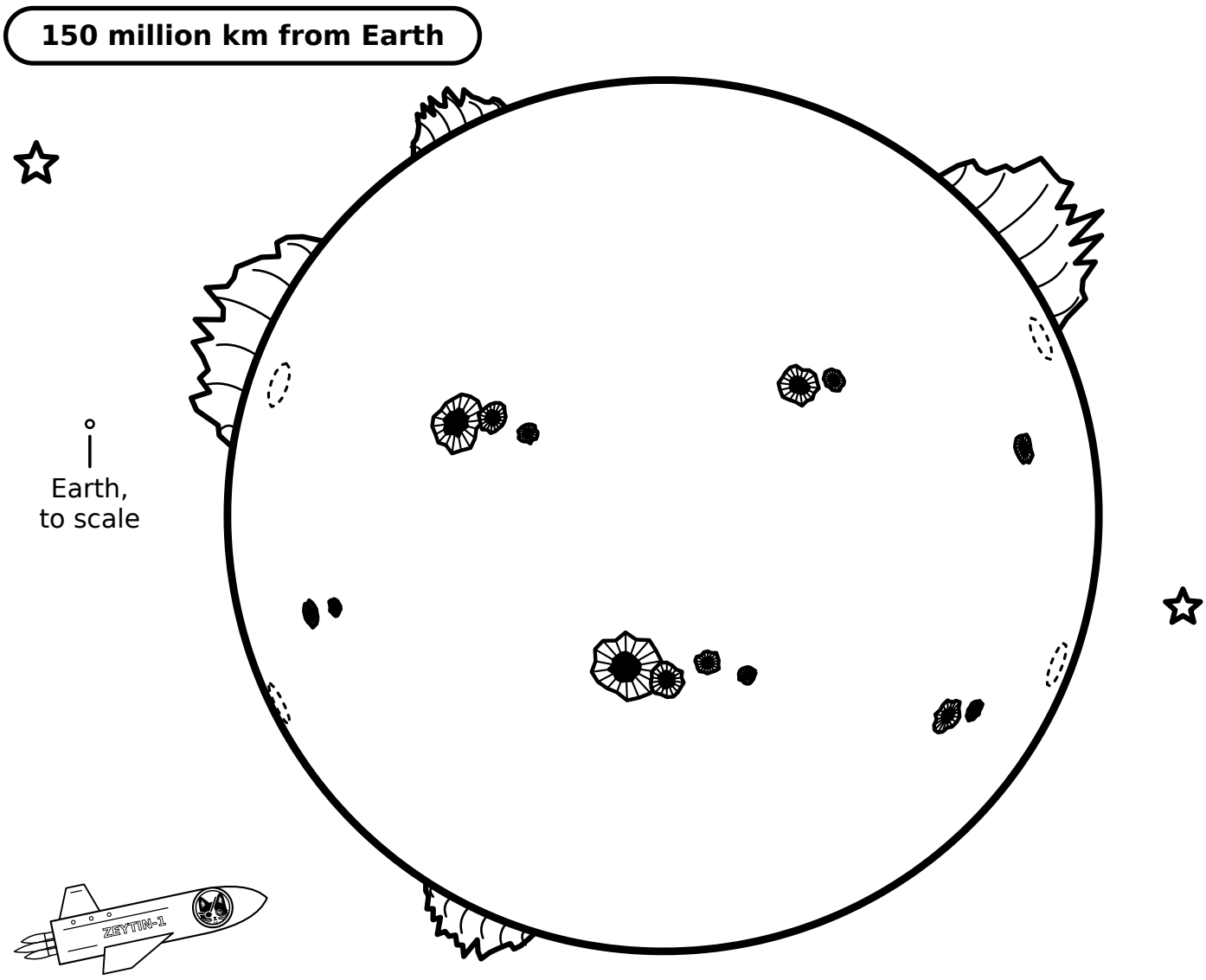
Footprints on the Moon

Next stop: the Moon, about 384,000 kilometers away — a three-day trip, just like the Apollo astronauts made.

Its dark patches are called seas, but they hold no water: they are plains of old lava that cooled more than three billion years ago.

With no wind or rain to wipe them away, the footprints astronauts left in 1969 can stay in the dust for millions of years!


ZEYTIN-1

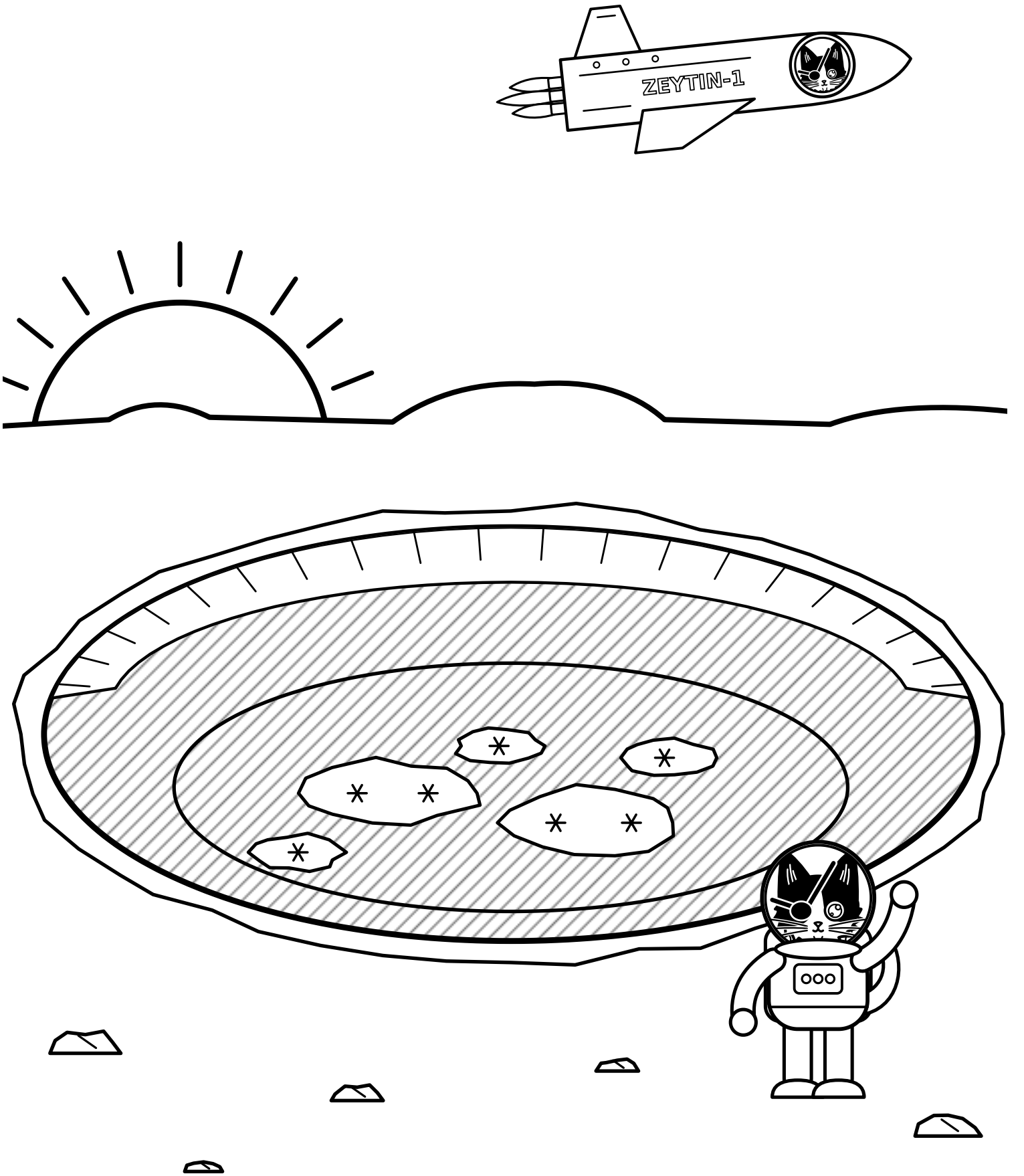


Our Star, the Sun

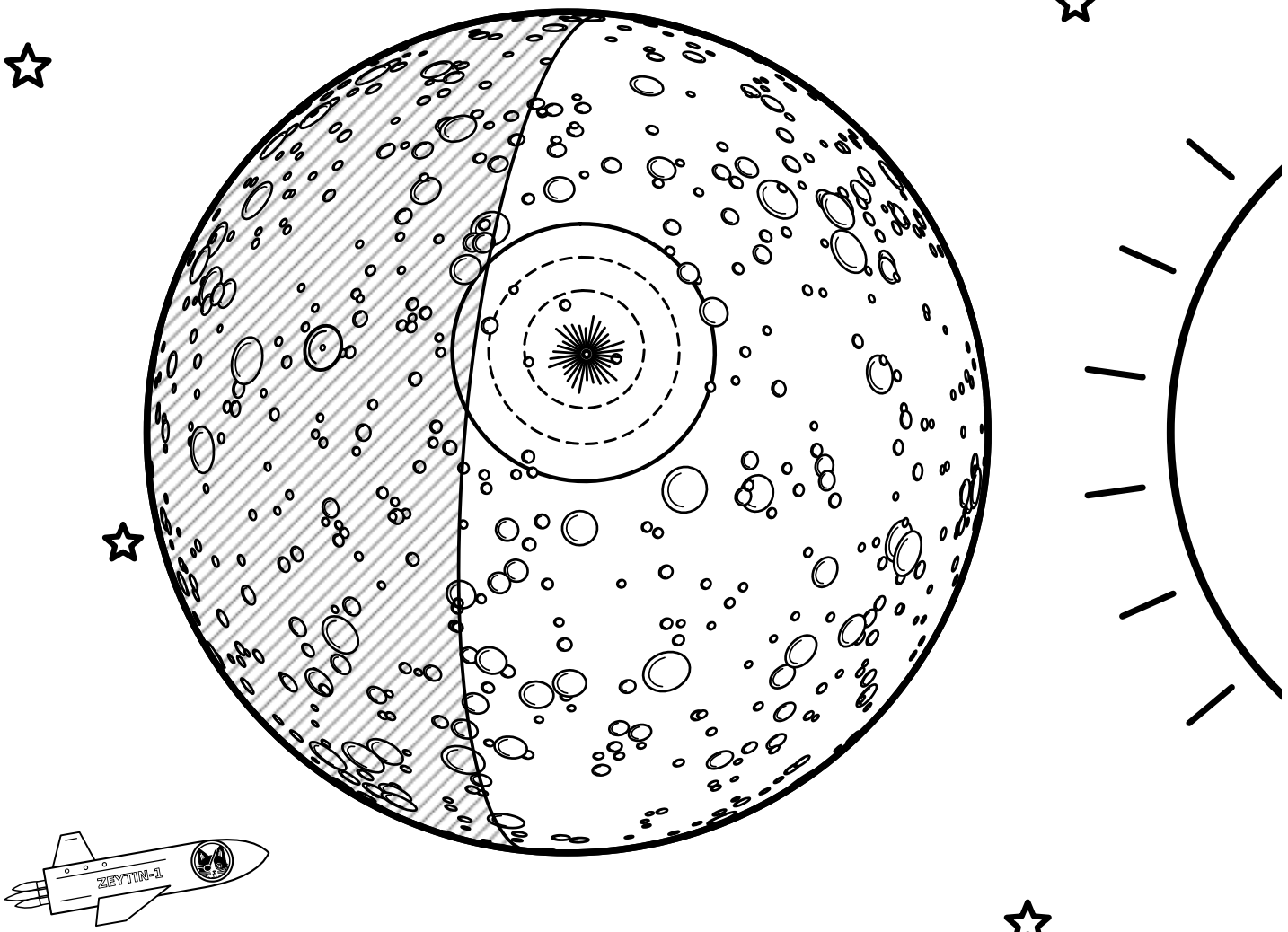
Zeytin flies toward the Sun, a giant ball of glowing gas so big that about 1.3 million Earths could fit inside it.

Its dark spots are sunspots: they look dark only because they are about 1,500 degrees cooler than the bright gas around them.

Above the spots, magnetic loops of hot plasma arch into space, so tall that many Earths could fit beneath them!


ZEYTIN-1

58 million km from the Sun

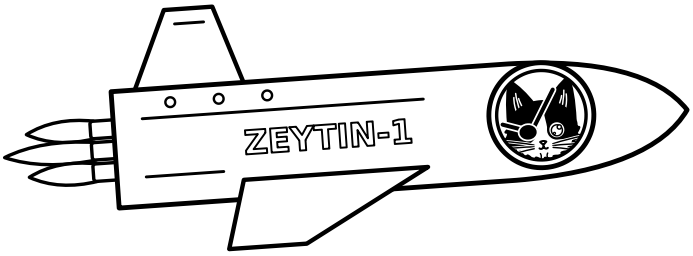


Mercury, Hot and Cold

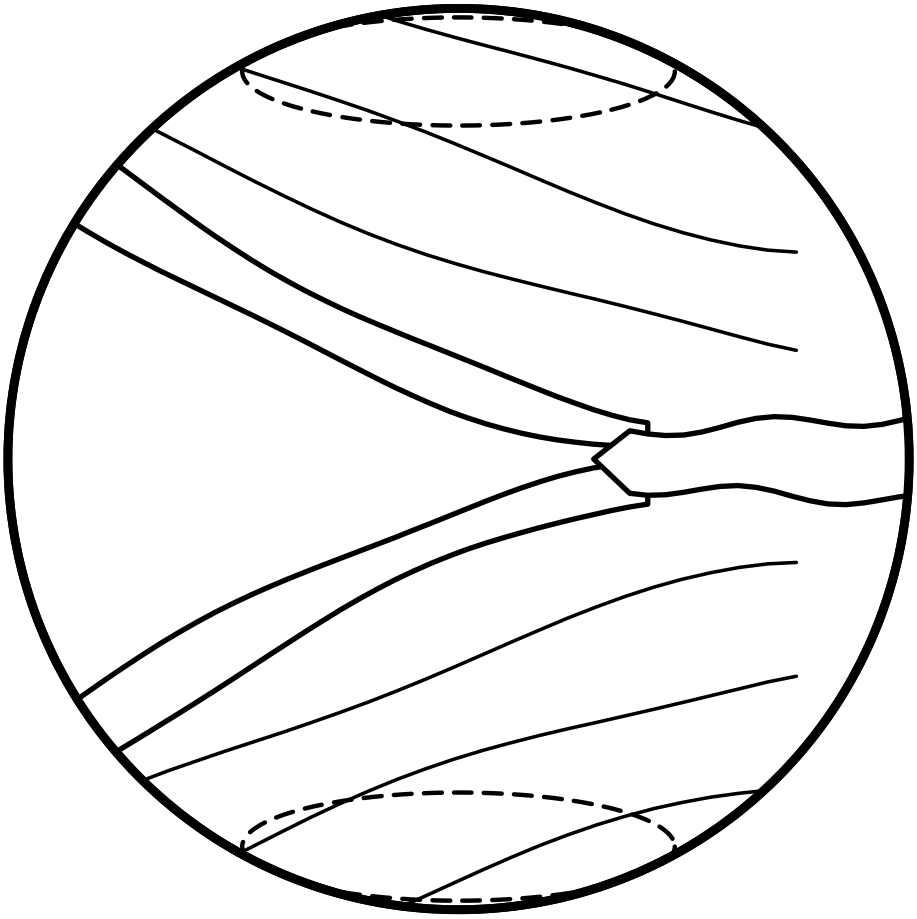
Mercury is the closest planet to the Sun, and one year there lasts only 88 Earth days.

In the daytime its ground heats up to about 430°C, hot enough to melt lead, but at night it drops to about −180°C.

Yet inside craters near its poles, where sunlight never reaches, scientists have found ice!


ZEYTIN-1

108 million km from the Sun



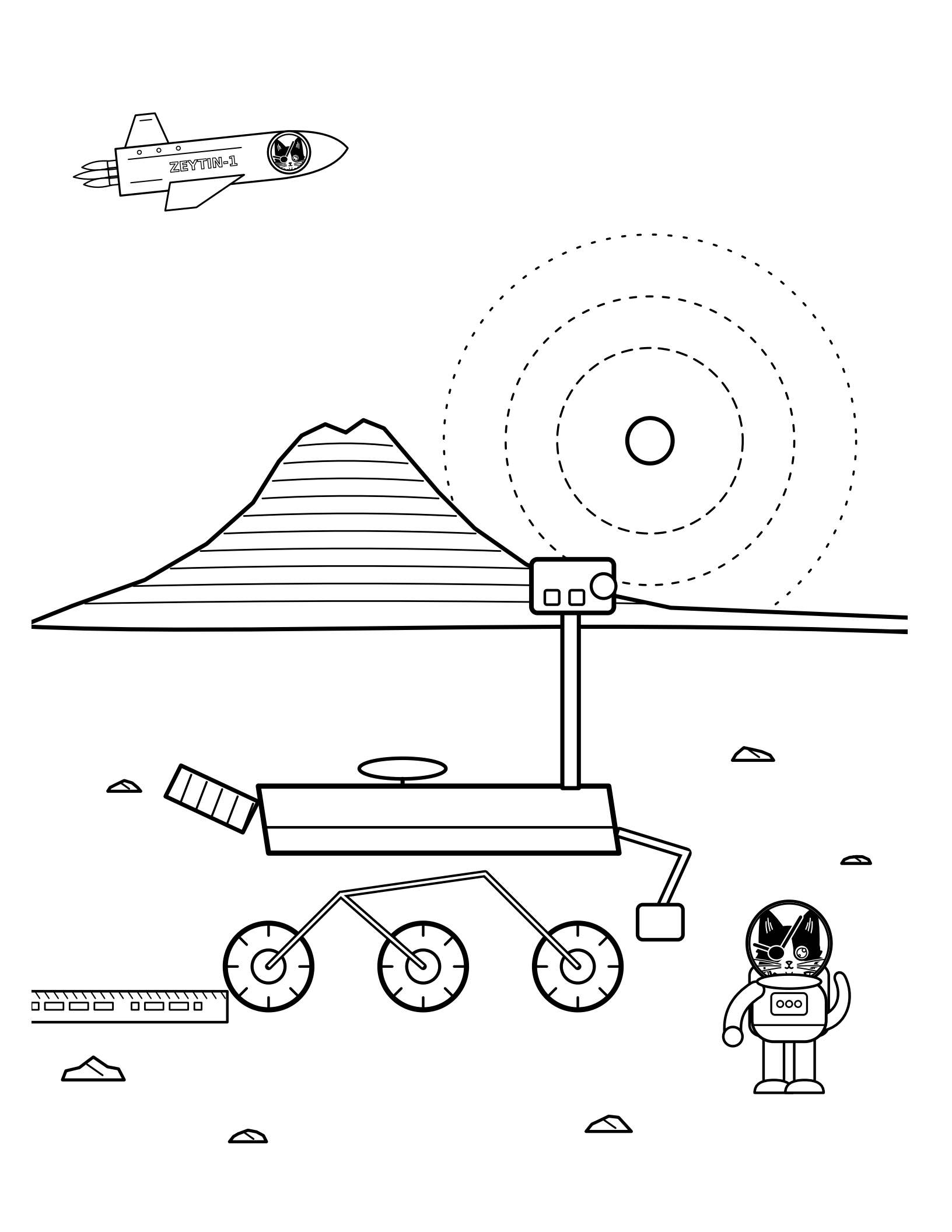


Venus, the Upside-Down Oven

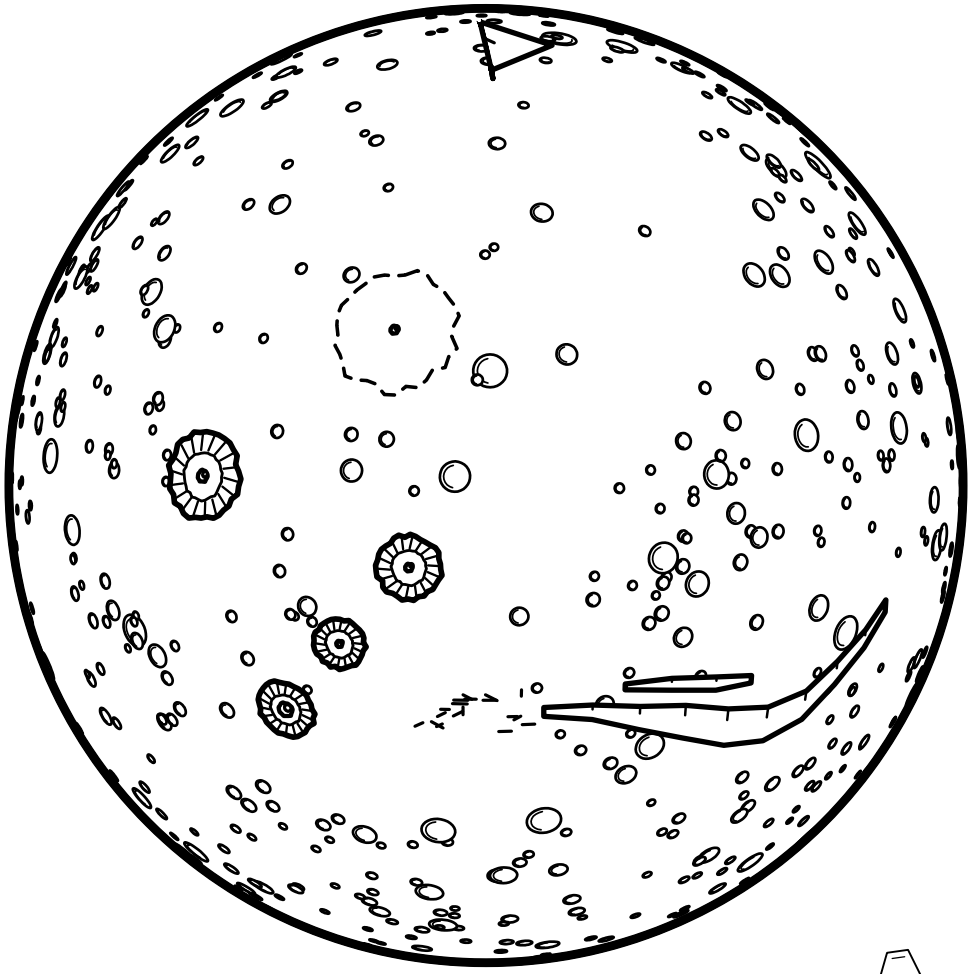
Venus hides under thick clouds of sulfuric acid, and its air is so heavy that standing there would feel like being 900 meters deep in the sea.

Its carbon dioxide air traps heat, so the ground is about 465°C — hotter than Mercury!

Venus also spins backward, and one slow turn takes 243 Earth days, longer than its 225-day year.


ZEYTIN-1

228 million km from the Sun



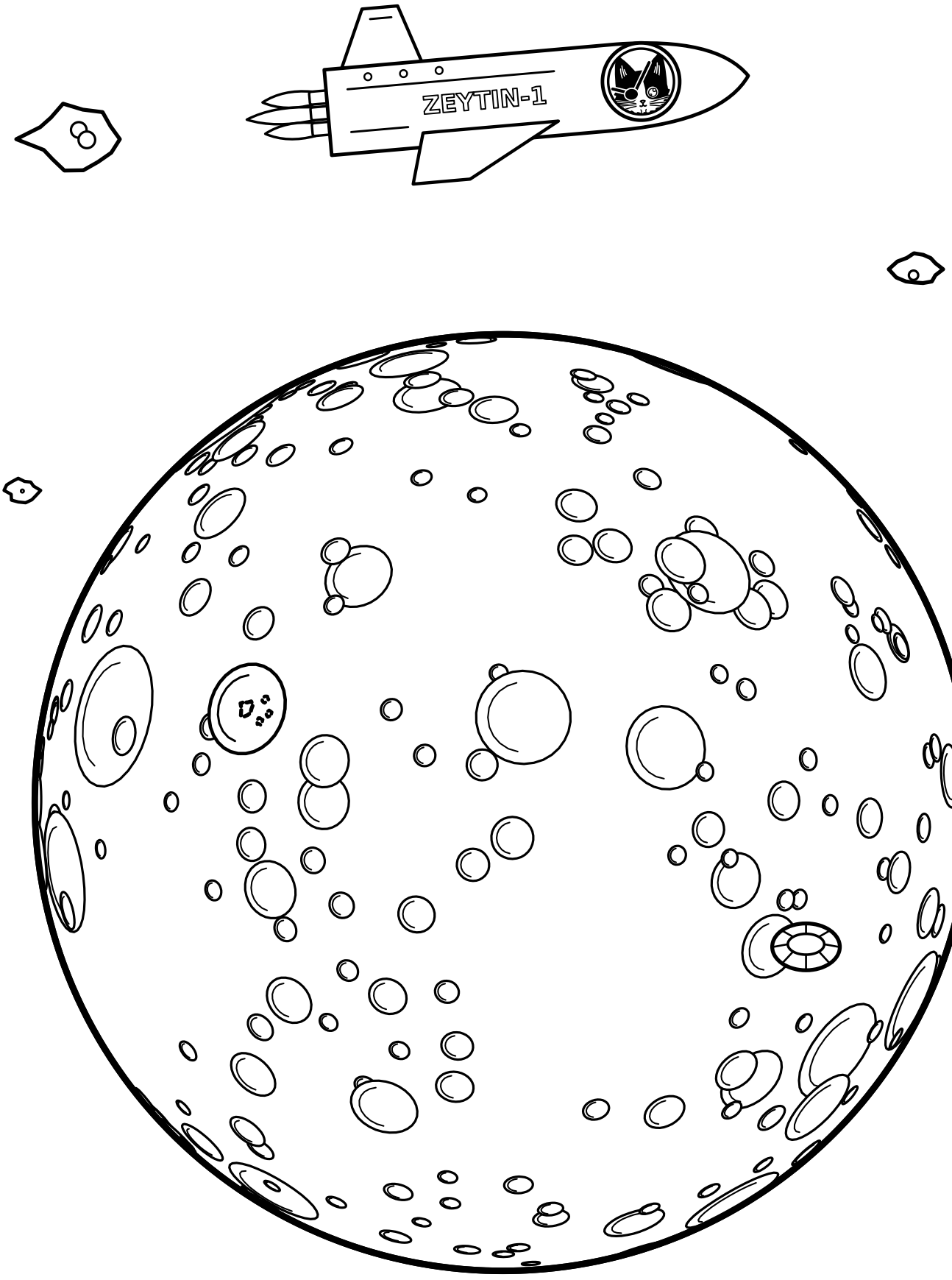


Mars, the Red Planet

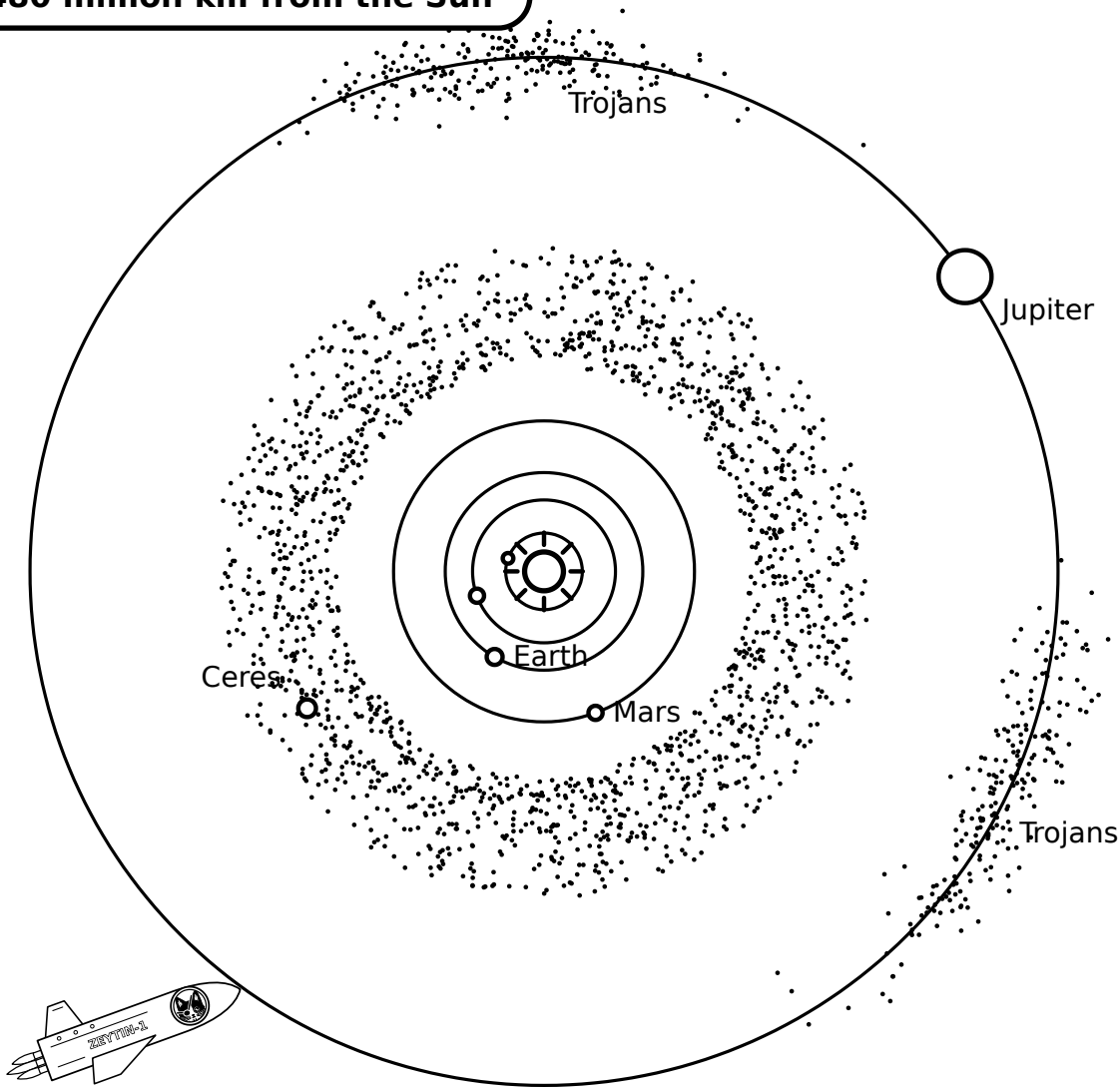
On Mars, Zeytin finds Olympus Mons, the tallest volcano in the Solar System: about 22 kilometers high, more than twice the height of Mount Everest.

The canyon Valles Marineris is about 4,000 kilometers long, nearly ten times longer than the Grand Canyon.

Rusty iron makes the dust red — and at sunset, the sky around the Sun turns blue!


ZEYTIN-1

330–480 million km from the Sun

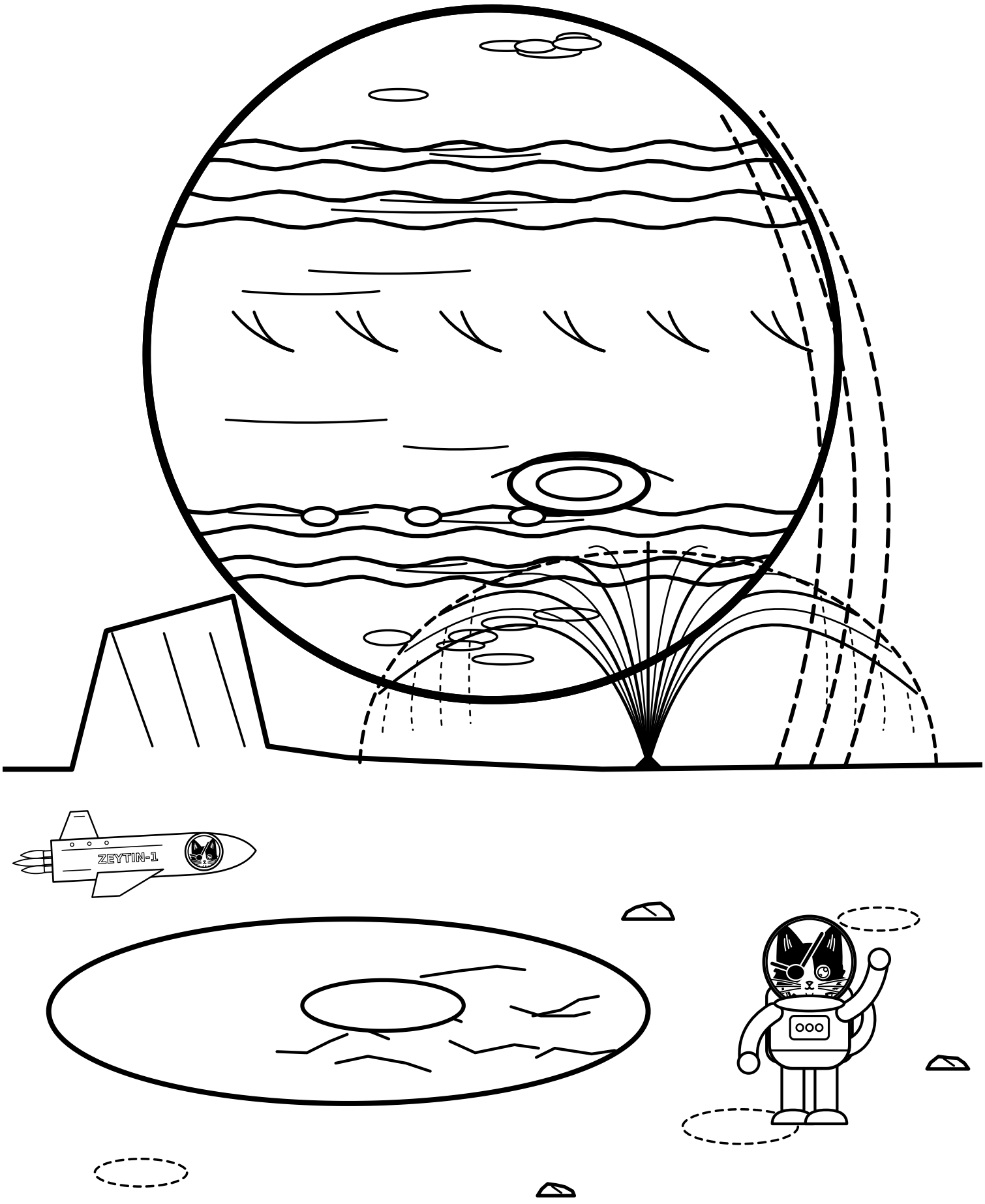


The Asteroid Belt

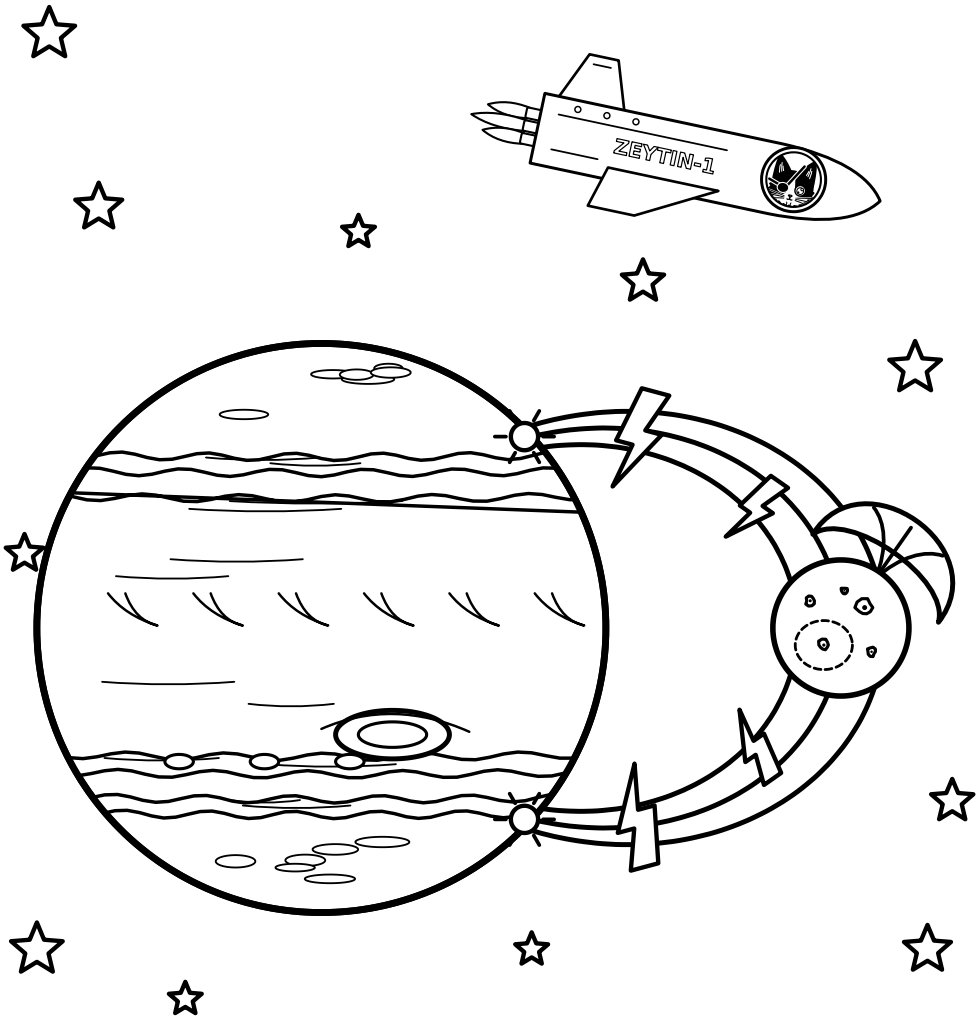
Between Mars and Jupiter lies the asteroid belt: millions of rocks left over from when the planets were born.

In movies they crowd together, but in real life they are so far apart that spacecraft fly through without hitting a single one.

The biggest, Ceres, is a dwarf planet about 940 kilometers wide, with bright salty spots in a crater called Occator.


ZEYTIN-1

778 million km from the Sun

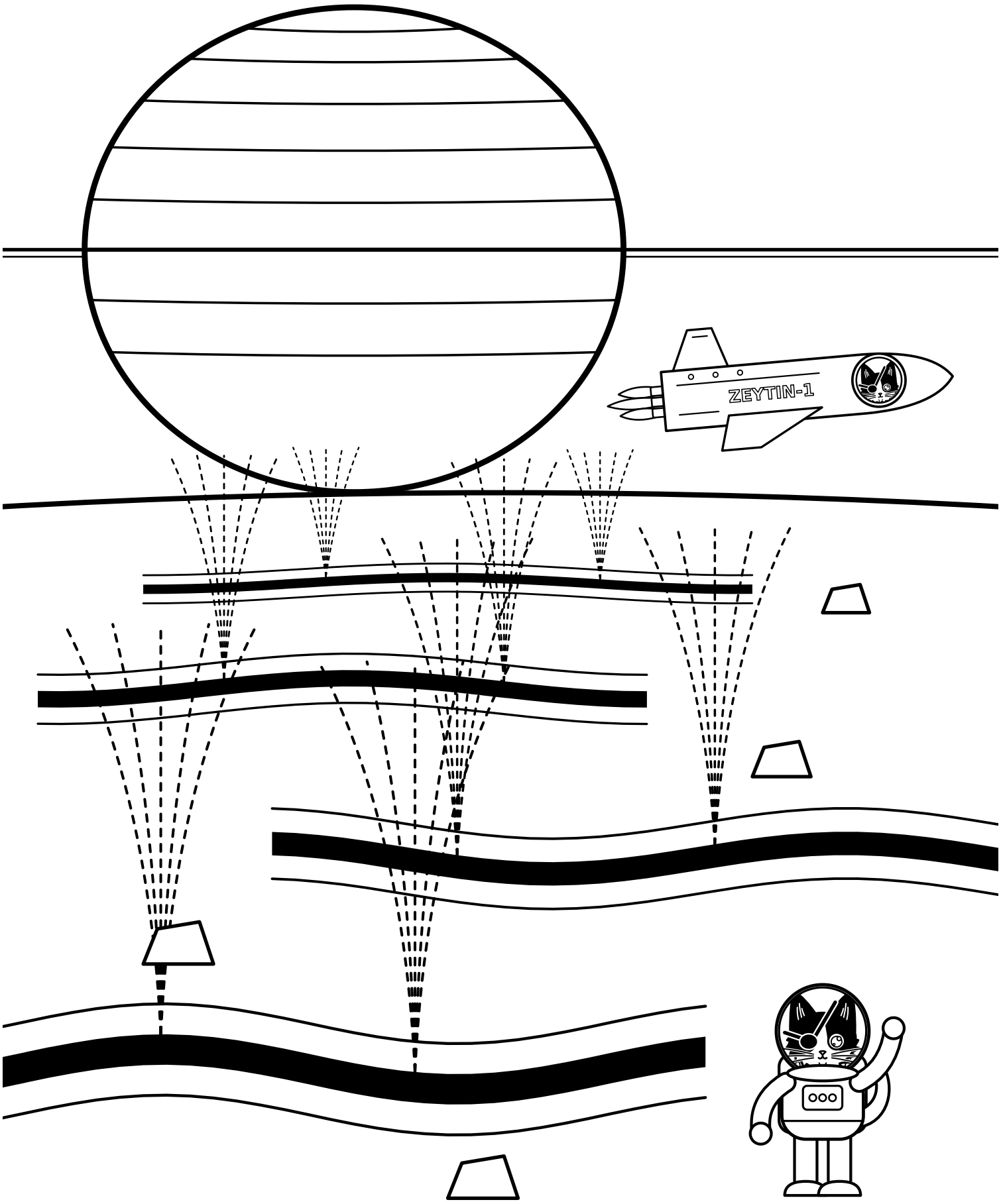


The Magnetic Dance

Every 42 hours, little Io dances once around giant Jupiter, always facing its partner and holding hands through an invisible magnetic bond.

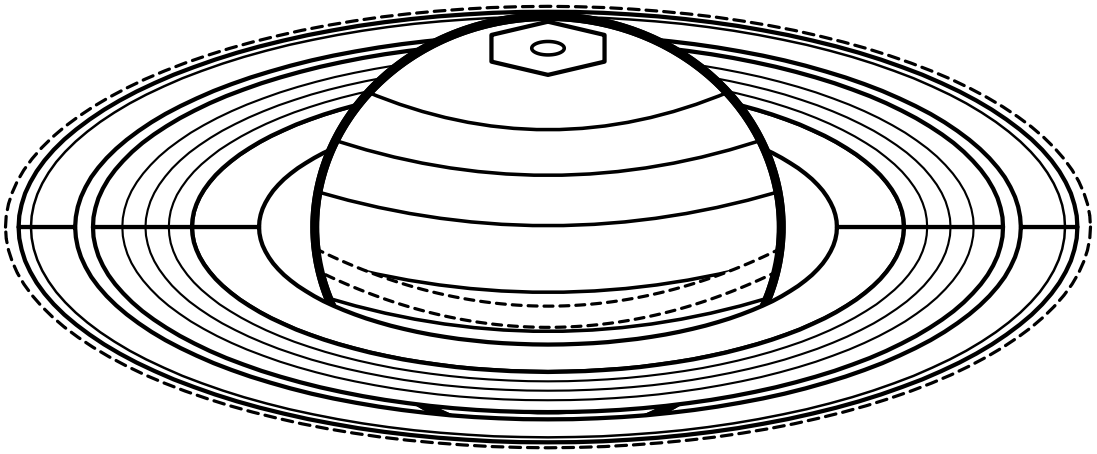
As Jupiter spins, its magnetic field sweeps past Io, and an electric current of about a million amperes flows along their bond like a sparkling ribbon.

Where the ribbon touches Jupiter's poles, glowing auroras light up — the shining footprints of Io's dance!

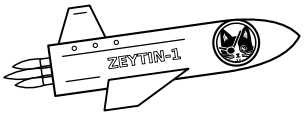

ZEYTIN-1

1.4 billion km from the Sun

Enceladus



Titan

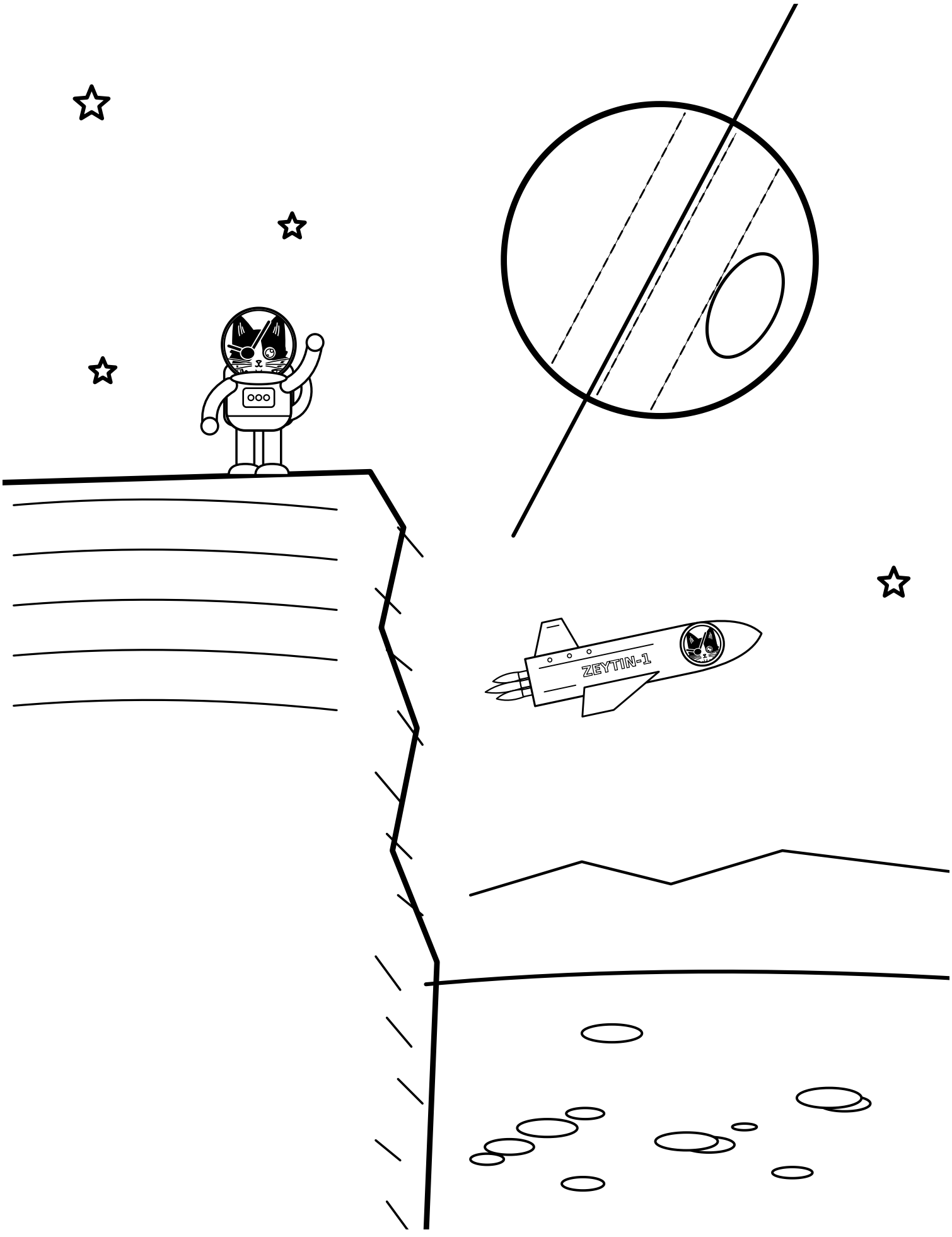


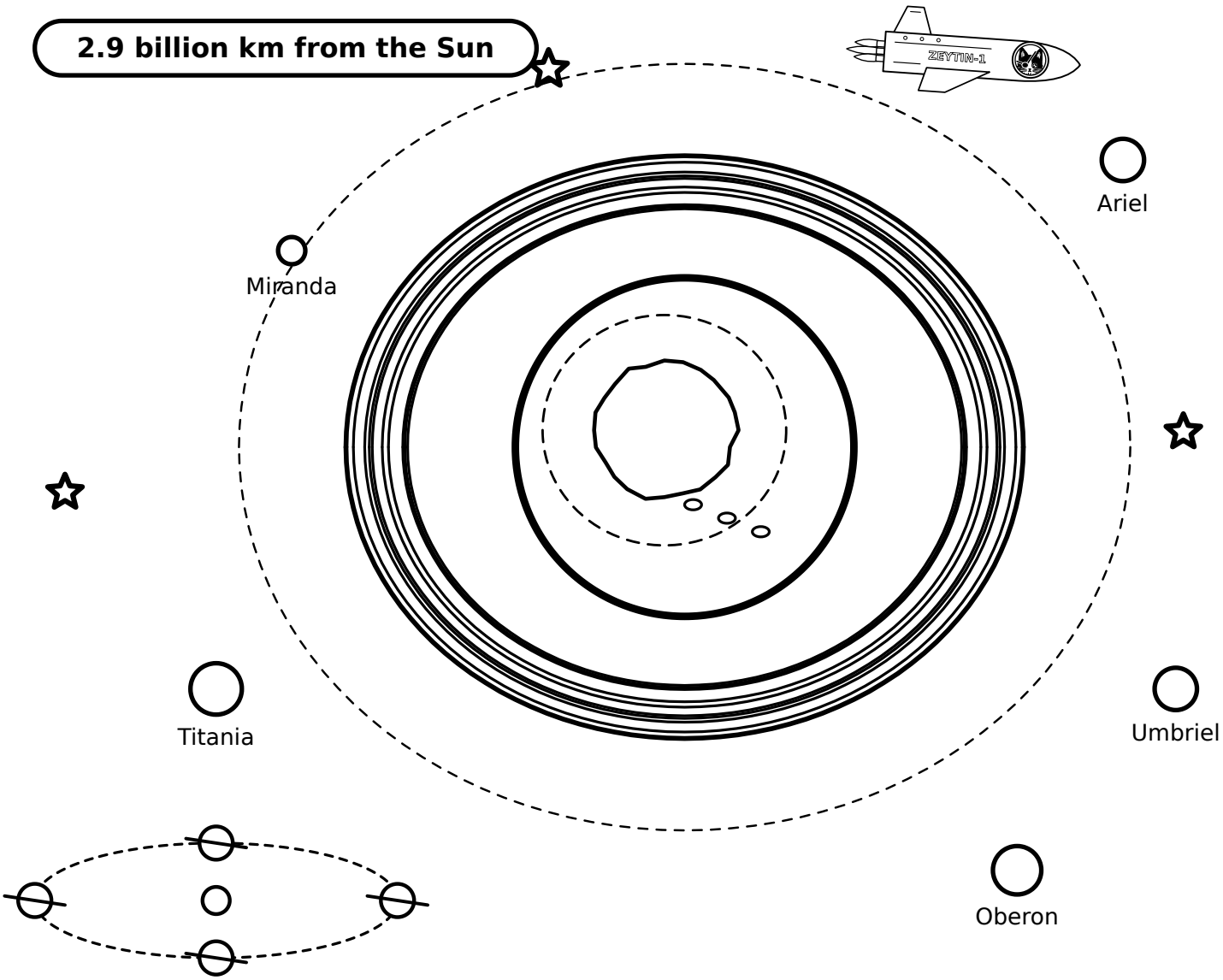
Saturn, the Floating Giant

Saturn is so light for its size that, in a bathtub big enough, it would float!

Its rings are made of countless pieces of ice, from tiny grains to chunks as big as a house, yet in most places they are only about ten meters thick.

Its little moon Enceladus shoots fountains of water into space from an ocean hidden under its icy shell.


ZEYTIN-1

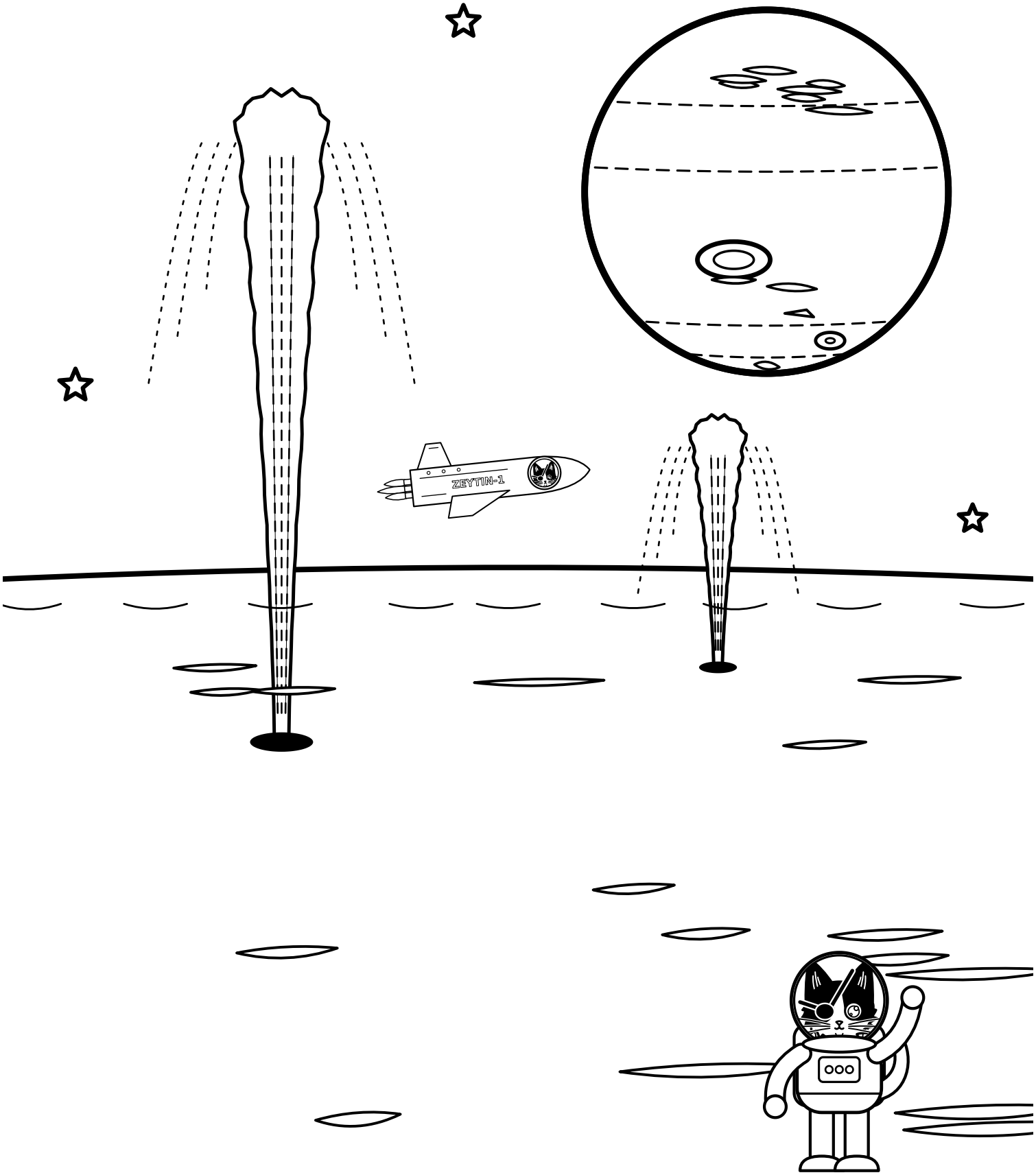


Uranus, the Sideways Planet

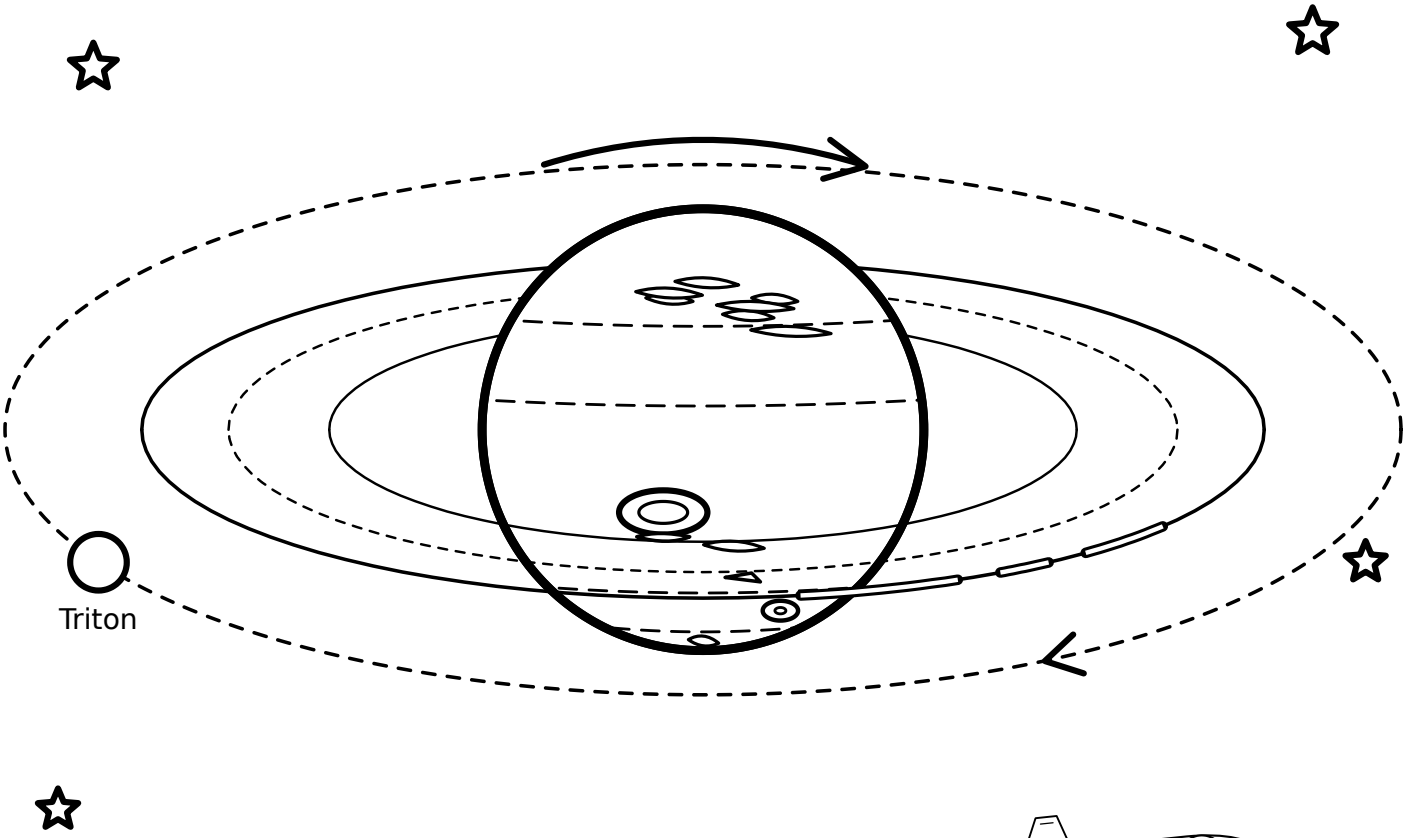
Uranus is tipped over on its side, so it rolls around the Sun like a ball.

Because of this, each pole gets about 42 years of sunshine, followed by 42 years of darkness.

Methane gas makes it blue-green, and it holds the record for the coldest air of any planet: about −224°C.


ZEYTIN-1

4.5 billion km from the Sun



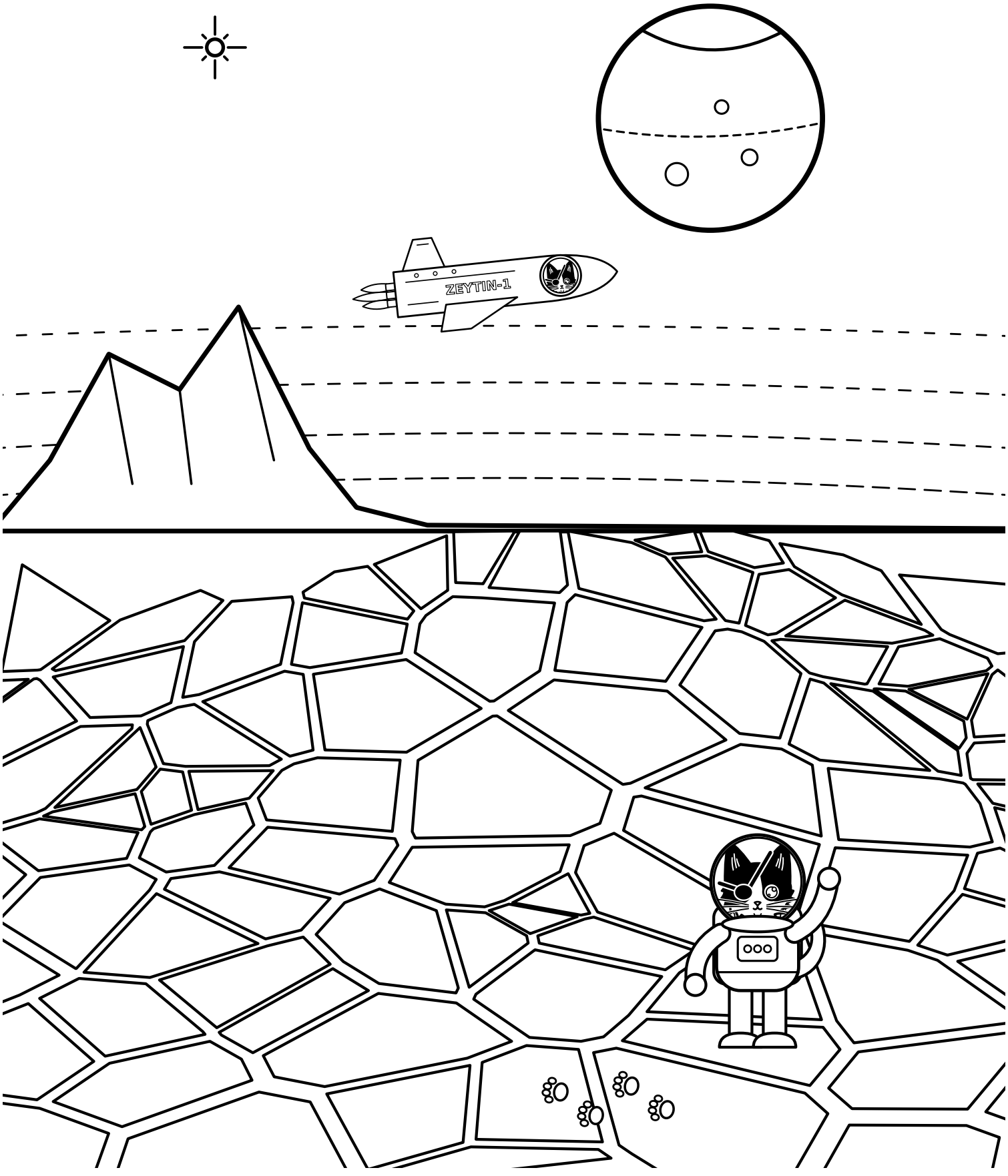


Neptune, the Windy World

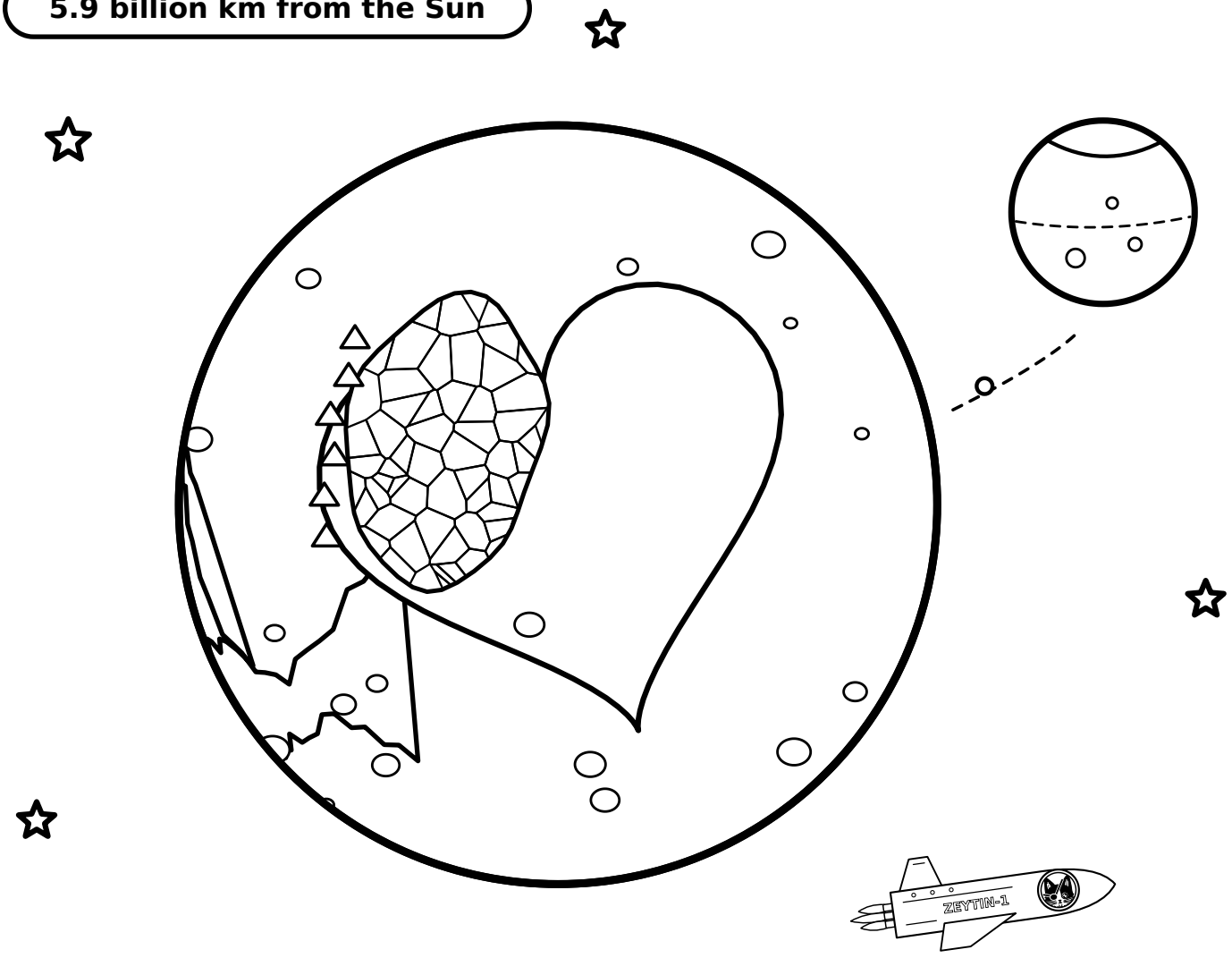
Neptune has the fastest winds in the Solar System, blowing at more than 2,000 kilometers per hour.

It was the first planet found with math: in 1846, astronomers worked out where it had to be — and there it was!

Its big moon Triton orbits backward, and its geysers shoot dark dust about 8 kilometers into the sky.


ZEYTIN-1

5.9 billion km from the Sun

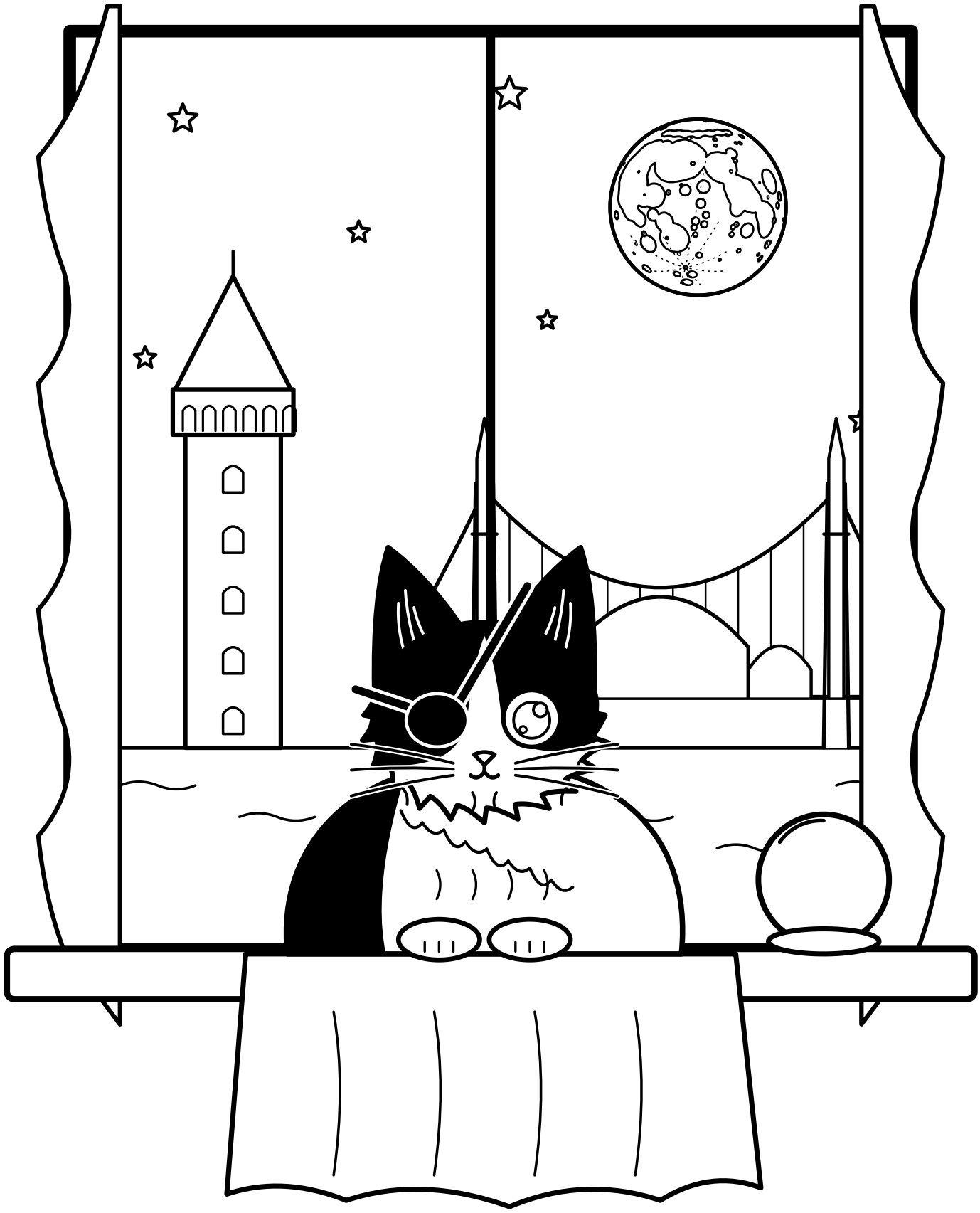


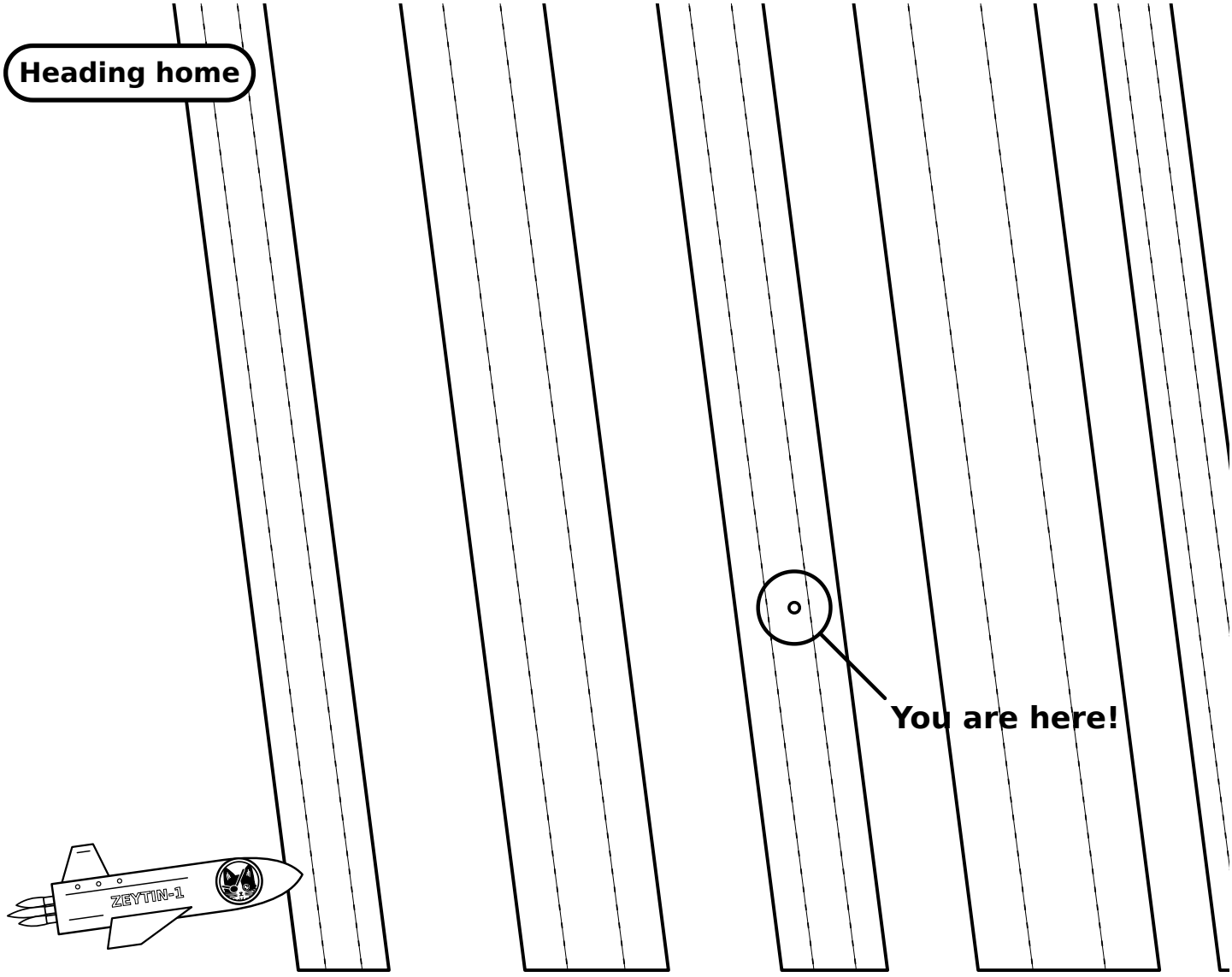
Pluto Has a Heart

Pluto is a dwarf planet so far away that sunlight takes about five and a half hours to reach it.

In 2015, the New Horizons spacecraft found a giant heart on Pluto: its left half is a plain of frozen nitrogen.

Its moon Charon is half as wide as Pluto, so the two spin around a point in space between them, like dancers holding hands.





The Pale Blue Dot

Before heading home, Zeytin turns around for one last look — and Earth is just a tiny, pale blue dot.

In 1990, the Voyager 1 spacecraft took a picture just like this from about 6 billion kilometers away.

Everyone who has ever lived is on that little dot. Let’s take good care of it!

Design Your Own Planet!



My planet is called _____

Is it hot or cold? _____

How many moons does it have? _____

Who lives there? _____